Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming 2nd mid-term test– December 21st 2023

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:15m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Could you please write a short pseudo code showing an example of a DOALL loop in C? Which parallel pattern is directly connected to the DOALL loop? (2)

B. Under what conditions can the following code be parallelized? If these conditions are met, is there a way to write the code for better parallelization? (2)

```
for (i=1; i<100; i++)
        for (j=0; j<10; j++)
            a[i][j] = f(a[j][i-1]);
```

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the scan pattern: assumptions, complexity, and example. (2)

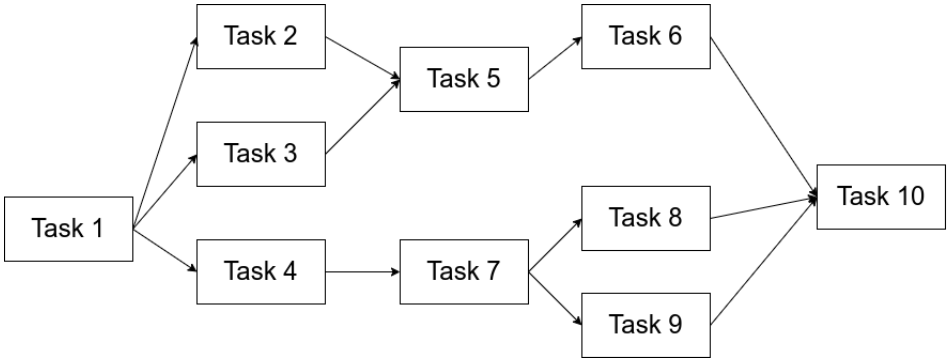
B. Please describe how the Split parallel pattern could be implemented. (2)

Exercise n. 3

Answer the following questions about shared memory parallel programming and briefly explain (without an explanation, the answer will be considered invalid)

A. Describe the main advantages of OpenMP over Pthreads. (1)

B. Consider the following task graph:



W(T1) = 100
W(T2) = 100
W(T3) = 100
W(T4) = 50
W(T5) = 100
W(T6) = 300
W(T7) = 50
W(T8) = 50
W(T9) = 50
W(T10) = 100

Calculate work and parallelism. Is it a good idea to use 3 threads? (1)
Write a corresponding OpenMP implementation. (2)

Exercise n. 4

Answer the following questions about MPI and briefly explain (without an explanation, the answer will be considered invalid)

A. Most of the point-to-point communication functions have a parameter named "tag". Please, state its type, explain its usage and why we might need to use the "MPI_ANY_TAG" value. (1) **This question can be skipped if you passed the challenge.**

B. Explain when the MPI usage overlaps with OpenMP and when OpenMP cannot replace MPI. (3)